LAN Network Management & Security

IT 221 | Public project summary (sanitized)

Overview

IT 221 group project: a small local area network (LAN) using a router and three laptops. The report describes static IP addresses, star topology, connectivity testing, remote access exploration, host firewall settings, and router MAC address filtering.

Network setup

The three laptops were connected through a router in a star topology. Static IP addresses were used to keep local addresses consistent.

Ping connectivity test

Ping commands were run against two other local devices (192.168.8.2 and 192.168.8.3). The report records successful replies, approximately 4–5 ms average response times, and no packet loss.

Remote access: TeamViewer

TeamViewer LAN mode was opened and a local device address was identified. The original report describes completing the remote connection as theoretically possible; it does not provide conclusive evidence that remote control was successfully completed. Connection credentials have been omitted from this public copy.

Windows Defender Firewall

The team explored Windows Defender Firewall settings. The original report describes creating an outbound blocking rule and testing it as theoretical steps, rather than documenting a completed application-blocking test.

Router MAC filtering

The team explored a Huawei router's MAC filtering configuration, including Allow/Deny modes, connected device listings, and a no-service screen described as a possible blocking consequence. Device MAC addresses and router-identifying screenshots have been omitted from this public copy. MAC filtering is a limited additional control because addresses can be spoofed.

Challenges

The team reported difficulty finding suitable cables and a working router, using an older router interface, and resolving overlapping remote-control screens and audio feedback.

Scope and evidence

This is a sanitized text-only public summary of the student project report. The original report contains screenshots, but these are excluded to avoid exposing identifiers, credentials, and other device information. This document does not independently verify hardware or security outcomes.

References from original report

A. S. Tanenbaum and D. J. Wetherall, Computer Networks, 5th ed., Pearson, 2011.

W. Stallings, Network Security Essentials: Applications and Standards, 6th ed., Pearson, 2017.

Microsoft TeamViewer listing (as cited in the original student report).